

# Ravi Gyani

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## PROFESSIONAL SUMMARY

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Hands-on Software Engineer with 20+ years of experience designing and operating cloud-native, event-driven systems at scale. Deep expertise in AWS, serverless architecture, and large-scale data processing. Consistent track record of cutting costs, eliminating technical debt, and shipping high-impact solutions across e-commerce, automotive, media, and national security domains.

## TECHNICAL SKILLS

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**AWS:** Serverless & compute (Lambda, ECS, Batch, Step Functions, API Gateway) | Data & analytics (S3, Glue, Athena, OpenSearch, Kinesis) | AI/ML (Bedrock, SageMaker) | IoT & edge (IoT Core, Greengrass) | Security (Inspector, IAM, Cognito)

**Infrastructure as Code:** Terraform, AWS CDK, CodePipeline — full lifecycle from development to production

**Big Data & Streaming:** Kafka, Spark, Flink, Airflow, Hadoop, HBase, Redis

**Containers & Orchestration:** Docker, Kubernetes (KEDA), Prometheus, Grafana

**Languages:** Python, Java, Golang

## CAREER HIGHLIGHTS

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### Delivered AI-First Search Experience — Tradebyte (Zalando)

- Designed an NLP-driven interface for OpenSearch, replacing traditional GUI filters with natural language queries, improving operator efficiency significantly.
- Fine-tuned a seq2seq model on Claude 3-generated training data, deployed via auto-scaling ECS for low-latency inference. Automated the full model lifecycle — dataset generation (Bedrock/S3), fine-tuning, containerization, and deployment — enabling rapid iteration as schemas evolve.

### Replaced Monolith with Event-Driven Microservices — Tradebyte (Zalando)

- Rearchitected a legacy monolith to a cloud-native, event-driven architecture on AWS Kinesis, DynamoDB, and Kubernetes (KEDA) enabling real-time data processing and fault isolation.
- Cut stock recalculation latency from 30 minutes to under 2 seconds using AWS Kinesis and Pandas.
- Consolidated fragmented upstream article-status streams into a single authoritative source of truth, reducing operational costs by 75% (\$12k → \$3k/month).
- When upstream services dropped events mid-launch, designed and shipped a reconciliation processor using S3 snapshots that reduced a potential week of manual recovery to a few hours.

### Eliminated a Database Dependency with Stateless RBAC - Luxoft GmbH

- Replaced a DB-backed permission system with Cognito custom attributes embedded in the JWT — removing an entire API Gateway + database round-trip while maintaining tamper-proof security.

### Reduced AMI Provisioning from 1 Day to 5 Minutes - Luxoft GmbH

- Automated Windows EC2 provisioning via Step Functions — cutting a full-day manual process to under 5 minutes, with auto-shutdown on CPU idle for cost savings.
- Implemented SSM Managed Instance Core, eliminating the need for public IPs entirely - turning a provisioning tool into a secure, self-service platform adopted across multiple teams.

### Built Federated ML Pipeline for Automotive Fault Detection - Luxoft GmbH

- Implemented Flower federated learning with AWS IoT Core, ECS, and Greengrass to train models across distributed vehicle telemetry without centralizing sensitive data, enabling real-time fleet-wide fault detection.

### **Slashed Cloud Compute Costs by 90% - Luxoft GmbH**

- Inherited a costly batch-processing workflow using Airflow, EFS, S3, and AWS Batch.
- Re-architected using Step Functions, Spot Instances, and S3 runtime mounts, cutting compute costs by 90% while improving reliability and scalability.

### **Eliminated 99% of Container CVEs via Automated Pipeline - Luxoft GmbH**

- Designed an automated ECR image rebuild and promotion pipeline: AWS Inspector → EventBridge → Step Functions → CodePipeline, with automated testing, output comparison, notification, and safe production promotion.

### **Scaled National Security Analytics to Billions of Location Records - Polaris Wireless**

- Replaced a failing SQL Server-based location tracking system with a custom big data platform (Hadoop, HBase, Kafka, Spark), delivering real-time subscriber profiling at national scale.

## **WORK EXPERIENCE**

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### **Principal Engineer — Tradebyte (Zalando), Munich Jan 2025 – Present**

- Architecting the migration from a tightly coupled monolith to a distributed, event-driven system, unlocking faster feature delivery and improved fault isolation.
- Leading AWS cloud adoption: spearheading POCs, migration blueprints, and aligning technology choices with business cost and performance objectives.
- Shaping strategic technical direction, balancing long-term scalability with near-term business growth.

### **Senior Solutions Architect — Luxoft GmbH, Munich Jan 2021 – Jul 2024**

- Embedded consultant across Amazon, BMW, Volkswagen, and Continental, owning end-to-end cloud architecture from design through delivery.
- Covered the full spectrum: migration strategy (BMW connected vehicle platform), ML pipeline design (VW), cost optimization, security automation, and IaC (Continental).

### **Solutions Architect — Polaris Wireless, Bangalore Feb 2014 – Aug 2020**

- Architected a scalable location analytics platform for national security, leading the big data transformation from SQL Server to Hadoop, HBase, and Spark.
- Invented and patented a real-time subscriber profiling and location estimation system (US20230050646A1).

### **Software Engineer & Founder — Various / KNOCKS Solutions 2003 – 2014**

- Built iOS/Android, .NET WinForms, and C# applications across multiple client engagements.
- Founded KNOCKS Solutions (2006–2009); shipped a commercial .NET WinForms UI component suite including Outlook-style calendars, VS-style docking, and Office 2007 ribbon controls.

## **EDUCATION**

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**Bachelor of Engineering - Netaji Subhas Institute of Technology, Delhi | 1999 – 2003**

## **CERTIFICATIONS**

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- AWS Certified Solutions Architect – Professional (SAP-C01) | Oct 2020
- AWS Certified Security Specialty (SCS-C01) | Mar 2023
- Confluent Certified Developer for Apache Kafka | Jul 2019
- Databricks Certified Developer: Apache Spark 2.X | Jul 2019

## **PATENT**

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### **Real-Time Subscriber Profiling & Location Estimation System - US20230050646A1**

Invented and filed during tenure at Polaris Wireless; enables scalable real-time analytics across billions of location records for national security applications.